Graphic Era Hill University, Bhimtal
Department of Computer Science and Engineering

B.Tech. CSE - Semester III - Academic Year 2026-27 (Odd Semester)
For *Sections D, E & F*

Unit I - Assignment

Course Code: TCS349

Course Name: Responsible and Explainable AI

Unit: Unit I - Introduction to AI Ethics, Bias and Fairness

Faculty: Mr. Manoj Kaushik, Assistant Professor, Department of CSE

Instructions:
Deadline is 8-August-2026.
Submit in A-4 size papers only (strictly).
Other type of paper pages are not acceptable.
The assignment will be discussed in class; do the assignment strictly by yourself for the best understanding.

Q1. Explain the importance of ethical AI development. Describe any two ethical dilemmas commonly faced during the design or deployment of an AI system.

Q2. Differentiate between ethics, morality and law as they apply to the development of AI systems.

Q3. Describe the different types of bias found in AI systems - data bias, algorithmic bias and interactional bias - giving one example of each.

Q4. Explain how feedback loops and deployment bias can affect the fairness of an AI system over time.

Q5. What is subgroup analysis? Explain its role in detecting bias in an AI model's predictions.

Q6. Using a confusion matrix, define True Positive Rate (TPR), False Positive Rate (FPR) and False Negative Rate (FNR). Explain how these measures help detect bias.

Q7. Define group fairness and individual fairness. How do demographic parity, equal opportunity and equalized odds differ from one another?

Q8. Explain the concept of predictive parity. Why is it generally not possible to satisfy equalized odds and predictive parity at the same time?

Q9. Describe pre-processing, in-processing and post-processing bias-mitigation strategies, giving one example technique for each.

Q10. Discuss the imminent risks and the long-term risks associated with the deployment of AI models in society.